

4-BIT ALU — RTL TO GDSII

Implementation using Verilog, OpenLane and Sky130

PROJECT REPORT

Design: 4-bit Combinational Arithmetic Logic Unit

Technology: Sky130

Physical Design Flow: OpenLane v1.0.2

RTL Design • Simulation • Synthesis • Floorplanning • Placement • Routing • Signoff

1. Abstract

This project implements a 4-bit combinational Arithmetic Logic Unit (ALU) using Verilog HDL and takes the design through a complete RTL-to-GDSII physical-design flow. The ALU supports four operations: addition, subtraction, bitwise AND and bitwise OR. RTL functionality was verified through simulation and waveform analysis. The design was synthesized and physically implemented using OpenLane with the Sky130 technology. The final flow completed routing, timing analysis and signoff checks, producing a valid GDSII layout.

2. Objectives

- Design a compact 4-bit combinational ALU using Verilog HDL.
- Verify all supported ALU operations through RTL simulation.
- Perform synthesis using an ASIC-oriented flow.
- Implement floorplanning, IO placement, placement and routing.
- Analyze timing and power results.
- Perform DRC and LVS signoff checks.
- Generate final GDSII, LEF and SPICE outputs.

3. ALU Architecture

The ALU accepts two 4-bit operands A and B and a 2-bit ALU_Sel control signal. The selected operation produces a 4-bit output Y. Because the design is combinational, there is no clock input.

ALU_Sel	Operation	Function
00	ADD	$Y = A + B$
01	SUB	$Y = A - B$
10	AND	$Y = A \& B$
11	OR	$Y = A B$

4. RTL Design

The ALU was described using a Verilog always block with a case statement. The ALU_Sel input determines which arithmetic or bitwise operation is applied to A and B. The source RTL is available in rtl/alu_4bit.v in the project repository.

5. RTL Simulation and Verification

The testbench applies representative operand and control combinations for all four ALU operations. Simulation was performed using Icarus Verilog and the resulting VCD waveform was inspected using GTKWave. The observed results included A=5, B=3: ADD produced 8, SUB produced 2, AND produced 1 and OR produced 7.

6. RTL-to-GDSII Flow

The physical-design flow was performed using OpenLane v1.0.2 and the Sky130 PDK. The major stages were synthesis, floorplanning, IO placement, power planning, global placement, placement optimization, detailed placement, global routing, detailed routing, timing analysis and signoff.

RTL	Verilog ALU
Simulation	Icarus Verilog / GTKWave

Synthesis	Logic synthesis
Floorplan	Core and IO planning
Placement	Standard-cell placement
Routing	Global and detailed routing
Signoff	STA / DRC / LVS / GDSII

7. Physical Design Results

Parameter	Result
Technology	Sky130
OpenLane	v1.0.2
Standard cells	34
Total cell area	335.3216 μm^2
TNS	0.00
WNS	0.00
Worst setup slack	4.06
Worst hold slack	4.00

8. Power Analysis

Power metric	Result
Total power	2.07×10^{-4} W
Internal power	9.80×10^{-4} W
Switching power	1.09×10^{-4} W
Leakage power	2.01×10^{-4} W

9. Clocking and CTS

This ALU is a purely combinational design and therefore has no clock port. Consequently, clock-tree synthesis (CTS) was disabled and is not applicable to this design. Timing analysis was still performed for the combinational paths.

10. Signoff Verification

The completed flow reported zero DRC violations and zero LVS errors. The final GDSII was generated successfully. The final design outputs include GDSII, LEF and SPICE representations.

Check	Result
DRC violations	0
LVS errors	0
Final GDSII	Generated successfully

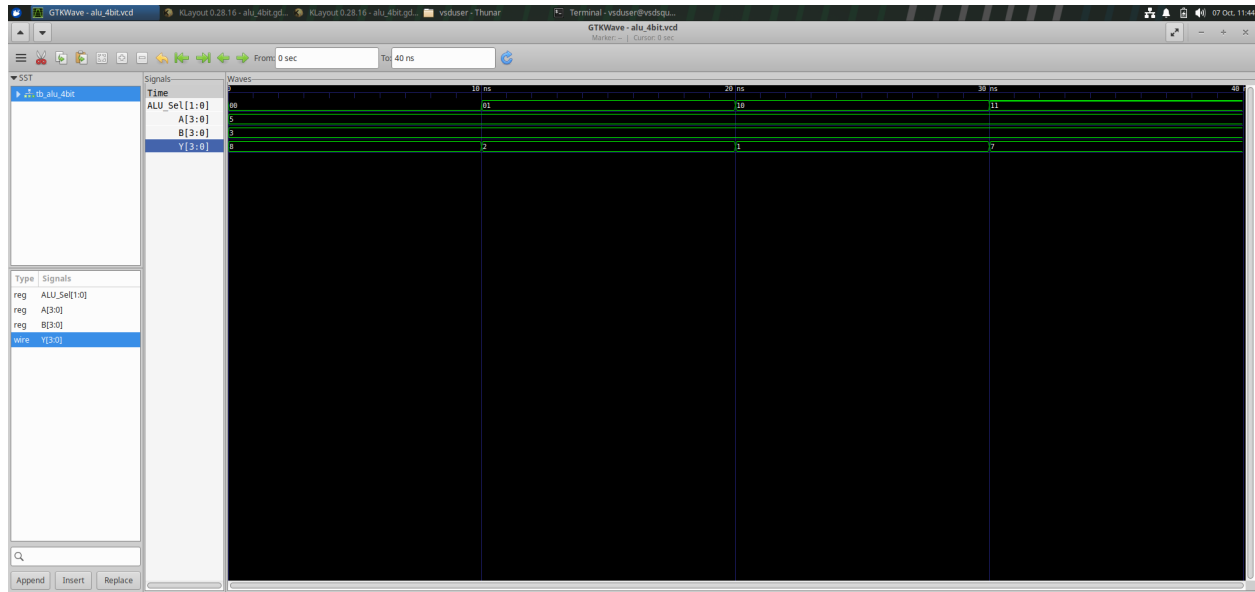
11. Final GDSII

The final physical layout was streamed to GDSII and successfully opened for inspection using KLayout. The repository contains results/final/alu_4bit.gds along with the corresponding LEF and SPICE files.

Simulation Evidence

```
vsduser@vsdsquadron:~/alu_project$  
File Edit View Terminal Tabs Help  
Terminal - vsduser@vsdsquadron:~/alu_project  
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the  
GNU General Public License for more details.  
  
You should have received a copy of the GNU General Public License along  
with this program; if not, write to the Free Software Foundation, Inc.,  
51 Franklin Street, Fifth Floor, Boston, MA 02110-1301, USA.  
  
FLAGS DLL vvp.tgt  
vvp.tgt: Icarus Verilog VVP Code Generator 12.0 (stable) ()  
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This program is free software; you can redistribute it and/or modify  
it under the terms of the GNU General Public License as published by  
the Free Software Foundation; either version 2 of the License, or  
(at your option) any later version.  
  
This program is distributed in the hope that it will be useful,  
but WITHOUT ANY WARRANTY; without even the implied warranty of  
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the  
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You should have received a copy of the GNU General Public License along  
with this program; if not, write to the Free Software Foundation, Inc.,  
51 Franklin Street, Fifth Floor, Boston, MA 02110-1301, USA.  
  
vsduser@vsdsquadron:~/alu_project$ yosys -V  
Yosys 0.47+121 (git sha1 90b4affc4, g++ 13.2.0-23ubuntu4 -fPIC -O3)  
vsduser@vsdsquadron:~/alu_project$ openroad -version  
openroad: command not found  
vsduser@vsdsquadron:~/alu_project$ nano alu_4bit.v  
vsduser@vsdsquadron:~/alu_project$ ls -l  
total 4  
rw-rw-r-- 1 vsduser vsduser 360 Oct  6 16:50 alu_4bit.v  
vsduser@vsdsquadron:~/alu_project$ nano tb_alu_4bit.v  
vsduser@vsdsquadron:~/alu_project$ nano alu_4bit.v  
vsduser@vsdsquadron:~/alu_project$ ls -l  
total 8  
rw-rw-r-- 1 vsduser vsduser 360 Oct  6 16:50 alu_4bit.v  
rw-rw-r-- 1 vsduser vsduser 816 Oct  6 16:54 tb_alu_4bit.v  
vsduser@vsdsquadron:~/alu_project$ iverilog -o alu_sim alu_4bit.v tb_alu_4bit.v  
vsduser@vsdsquadron:~/alu_project$ vvp alu_sim  
Time=0 | A= 5 | B= 3 | Sel=00 | Y= 6  
Time=10000 | A= 5 | B= 3 | Sel=01 | Y= 2  
Time=20000 | A= 5 | B= 3 | Sel=10 | Y= 1  
Time=30000 | A= 5 | B= 3 | Sel=11 | Y= 7  
tb_alu_4bit.v:43: $finish called at 40000 (1ps)  
vsduser@vsdsquadron:~/alu_project$
```

Waveform Evidence



12. Key Learning Outcomes

- Verilog RTL design of combinational digital logic.
- RTL simulation and waveform-based verification.
- ASIC synthesis and standard-cell implementation.
- Understanding of floorplanning, placement and routing.
- Timing and power analysis.
- Physical verification using DRC and LVS.
- Generation and inspection of a final GDSII layout.

13. Conclusion

The 4-bit ALU successfully progressed from Verilog RTL through simulation and the OpenLane physical-design flow to final GDSII generation using the Sky130 technology. The completed implementation contains 34 standard cells, reports zero TNS and WNS, has positive reported setup and hold slack, and passed DRC and LVS with zero violations/errors. The project demonstrates a complete introductory ASIC RTL-to-GDSII workflow.

14. Future Scope

Future work includes extending the ALU with XOR, NOT, shifts, and additional arithmetic operations.